# Ultrathin foils used for low-energy neutral atom imaging of the terrestrial magnetosphere

**Herbert O. Funsten**
**David J. McComas**
**Bruce L. Barraclough**
Los Alamos National Laboratory
Space and Atmospheric Sciences Group
MS D466
Los Alamos, New Mexico 87545

**Abstract.** Magnetospheric imaging by remote detection of low-energy neutral atoms (LENAs) that are created by charge exchange between magnetospheric plasma ions and neutral geocoronal atoms has been proposed as a method to provide global information of magnetospheric dynamics. For LENA detection, carbon foils can be implemented to (1) ionize the LENAs and electrostatically remove them from the large background of solar UV scattered by the geocorona to which LENA detectors (e.g., microchannel plates) are sensitive and (2) generate secondary electrons to provide coincidence and/or LENA trajectory information. Quantification of LENA-foil interactions are crucial in defining LENA imager performance. We present equilibrium charge state and angular scatter distributions for 1- to 30-keV atomic hydrogen, helium, and oxygen transiting nominal 0.5 $\mu$g cm$^{-2}$ carbon foils. Results obtained after coating the exit surface of foils with either aluminum or gold suggest that the intended alteration of the exit surface chemistry has no effect on the charge state distributions due to foil contamination from exposure to air. Angular scattering that results from the projectile-foil interaction is quantified and is shown to be independent of the charge state distribution.

*Subject terms: magnetospheric imagery; atmospheric remote sensing; neutral atom detection; charge exchange; beam-foil interactions.*

*Optical Engineering 32(12), 3090–3095 (December 1993).*

## 1 Introduction

The terrestrial magnetosphere has been extensively studied using *in situ* observations acquired along various spacecraft orbits. Unfortunately, interpretation of such single-point measurements is ambiguous because of dynamic variations of the magnetosphere. However, remote imaging of the magnetosphere has been proposed by detection of energetic neutral atoms[1] (ENAs) and low-energy neutral atoms[2] (LENAs) which are created by neutralization of magnetospheric ions (predominantly hydrogen, helium, and oxygen) by charge exchange with geocoronal neutral species. Because ENAs and LENAs do not interact with local electromagnetic fields, they travel in ballistic trajectories and can be detected remotely. A remote LENA detector would provide global information to map magnetospheric plasma source locations, strengths, and energy distributions.[3]

Paper MI-004 received July 15, 1992; revised manuscript received July 19, 1993; accepted for publication July 21, 1993. This paper is a revision of a paper presented at the SPIE conference on Instrumentation for Magnetospheric Imagery, July 1992, San Diego Calif. The paper presented there appears (unrefereed) in SPIE Proceedings Vol. 1744.


Because of the sensitivity of LENA detectors (e.g., microchannel plates) to ultraviolet light, detection of magnetospheric ENAs and LENAs requires their removal from intense photon fluxes of direct solar UV and solar UV scattered by the terrestrial geocorona. For ENA detection, UV can be filtered using a thick blocking foil (e.g., ~15 $\mu$g cm$^{-2}$ carbon),[4] which is opaque[5,6] to UV. However, because of their low energies, LENAs will not transit a thick blocking foil without severe energy straggling and angular scattering. LENAs can be removed from the ambient UV by first ionizing them and then electrostatically deflecting them into an energy-angle measurement section of the imager.[7] The sensitivity of the imager is directly proportional to the ionization efficiency of the charge-stripping mechanism.

A straightforward method of LENA charge modification utilizes ultrathin carbon foils for charge stripping.[7] Carbon foils are structurally robust and can withstand the severe conditions of launch and space. Ultrathin foils of nominal thickness 0.5 $\mu$g cm$^{-2}$ can be employed to minimize angular scattering and energy straggling of LENAs that can degrade instrument performance. In this paper, we present experimental results describing the equilibrium charge state and angular scatter distributions of low-energy (kilo-electron-

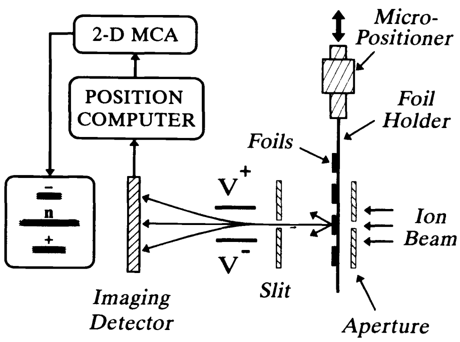



**Fig. 1** Experimental apparatus.

volt) atomic hydrogen, helium, and oxygen that transit ultrathin (nominal 0.5 $\mu g\ cm^{-2}$) foils. We demonstrate that coating of the surface with either gold or aluminum has no observable effect on the hydrogen charge state distributions.

## 2 Experiment

The experimental apparatus used in this study, which is illustrated in Fig. 1 and is similar to the apparatus used by Burgi et al.,[8] allows for simultaneous charge state and angular scattering analysis of a transmitted ion beam. Ions are generated in a duoplasmatron ion source, accelerated to an energy of 1 to 30 keV, and magnetically analyzed. The mass-selected ions pass through a 1-mm-diam beam-defining aperture located 12.5 mm upstream of a foil holder, which can hold five foils, each of which can be moved into the beam by a micropositioner.

After transiting the foil, the initially collimated ion beam is composed of scattered ions and neutrals. Some of these atoms pass through a 0.35- × 25-mm horizontal slit located 30 mm behind the foil. Because the center of the slit corresponds to the beam axis, the horizontal distribution of atoms contains direct information about the angular scattering of the atoms in the foil. Ions that pass through the slit are vertically energy-per-charge (*E/q*) analyzed by a pair of deflection plates. All atoms that pass through the slit subsequently strike an imaging microchannel plate (IMCP) detector located 70 mm behind the slit. Neutrals, which are not deflected, make an image of the slit across the center of the IMCP detector, whereas ionized species are deflected toward the top and bottom where they also form images of the slit on the IMCP detector. The vertical position of the negative, neutral, and positive charge state distributions is a function of the deflection voltage and *E/q* of the ions. Because energy loss of the projectiles is relatively small and only three charge states are typically observed, the exit charge states are well separated on the IMCP detector.

A position computer determines locations of projectiles detected by the IMCP detector. This information is then displayed using a 2-D multichannel analyzer (MCA). A background subtraction is performed on the integrated charge state distributions, which are then converted into a fraction relative to the other charge states. For example, the charge state distribution of an incident proton beam after transiting a foil is composed of fractions of negative ions $f(H^-)$, neutrals $f(H^0)$, and protons $f(H^+)$, where $f(H^-)+f(H^0)+f(H^+)=1$.

Ultrathin foils used in this study can have pinholes that will detrimentally affect analysis of the scatter and charge state distributions. Incident ions with charge state $q=+1$ that transit a hole are neither scattered nor charge modified and will therefore be observed as a localized high-count region in the $q=+1$ charge state distribution. Using the IMCP detector, the high-count region corresponding to a hole can be observed, and during subsequent analysis this region can easily be avoided. In fact, using a novel technique of transmitted ion mapping, structural flaws such as holes can be observed and optimal foils can be selected.[9]

The final charge state of atoms exiting a solid has previously been shown to be somewhat sensitive to the exit surface chemistry.[10–12] To investigate whether coatings affected the exit charge state of transiting atoms, Al and Au layers were deposited on the exit surface of the carbon foils. These layers, each 17-Å thick, were deposited by evaporation in a vacuum chamber at $6\times10^{-6}$ Torr. These foils were exposed to air overnight (on exposure to air, the Al layer should immediately form aluminum oxide) and were installed into the experimental chamber the following day. They were kept under vacuum at a pressure of $5\times10^{-8}$ Torr for several days prior to charge state and scattering experiments.

## 3 Charge State Distributions

As an atomic beam traverses a foil, the charge state composition of the beam changes because of electron loss and capture events associated with the interaction of the projectiles and the atoms in the foil. After a certain number of interactions, which corresponds to a penetration depth $\lambda$, an equilibrium charge state distribution of the beam is established. Extrapolating empirical charge equilibration depths derived at somewhat higher energies,[13] charge equilibration of an ion beam at the energies used in this study should occur within a penetration distance $\lambda<4$ Å in the carbon foil, which corresponds to a foil thickness of $\sim$0.1 $\mu g/cm^2$. At this depth, the beam retains no information of its initial charge state. Because $\lambda$ is much less than the thickness of the foils used in this study and charge equilibration has rapidly occurred, we can infer that our results using incident ions ($H^+$, $He^+$, and $O^+$) apply to incident neutrals. Additional empirical evidence supporting charge equilibration includes (1) the incidental exit charge state distributions obtained from incident beams of different charge states (e.g., $O^{1+}$, $O^{2+}$, $O^{7+}$) using 1 to 2 $\mu g\ cm^2$ carbon foils[14] and (2) the independence of the charge state distribution on angular scattering (see Sec. 4).

Equilibrium charge state distributions of incident $H^+$, $He^+$, and $O^+$ beams that transit nominal 0.5 $\mu g/cm^2$ carbon foils are depicted in Fig. 2 as a function of incident ion energy $E$. The solid lines in the figures represent least-squares fits of the data to the equations

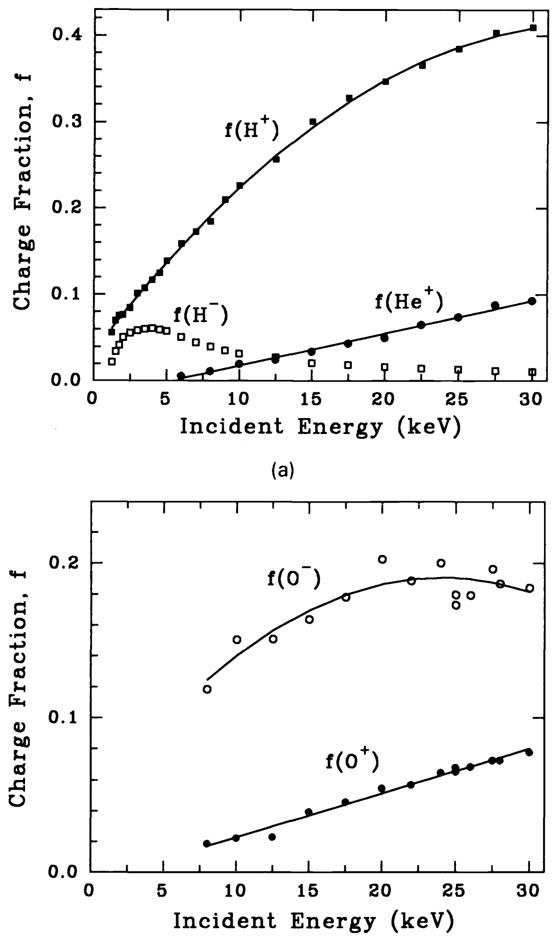
$$f(H^+)=-3.24\times10^{-4}E^2+0.0222E+0.0342\ , \qquad (1)$$

$$f(He^+)=3.75\times10^{-3}E-0.0197\ , \qquad (2)$$

$$f(O^+)=2.86\times10^{-3}E-5.79\times10^{-3}\ , \qquad (3)$$

$$f(O^-)=-2.58\times10^{-4}E^2+0.0124E+0.0418\ , \qquad (4)$$

where $E$ is in units of kilo-electron-volts. Because $\lambda$ is much less than the foil thickness, the exit charge state of an indi-




(a)

(b)

**Fig. 2** Equilibrium exit charge states for (a) He and H and (b) O transiting a nominal 0.5 $\mu$g cm$^{-2}$ carbon foil as a function of their incident energy. Solid lines correspond to Eqs. (1) through (4).

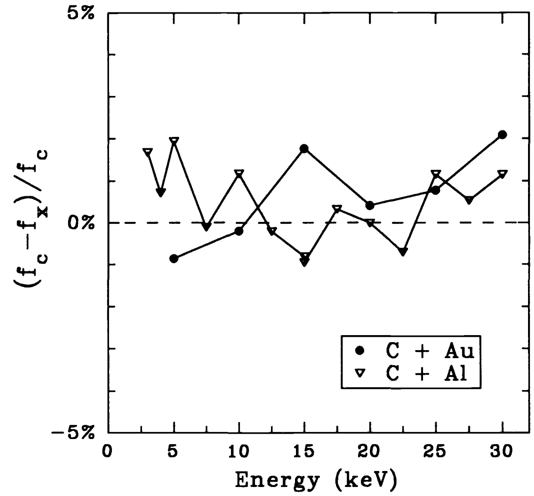



**Fig. 3** Variation (as defined in the text) of $f(\mathrm{H}^+)$ for carbon foils coated with Au or Al relative to that of an uncoated carbon foil.

vidual atom transiting the foil is determined at the exit surface and, therefore, at the atom's exit velocity. In Fig. 2, the exit charge state distributions are plotted as a function of incident ion energy, because energy loss in the foil is considered here to be an instrument calibration issue. To extend these results to foils of other thicknesses, energy loss in the foil must be considered[15] and can be calculated using the TRIM computer code.[16]

For hydrogen, the fraction of the beam exiting the foil as $\mathrm{H}^+$ increases with increasing energy, whereas the exit charge state fraction of $\mathrm{H}^-$ reaches a maximum at approximately $E=4$ keV and subsequently decreases with increasing energy. Based on counting statistics, the error is less than 0.5%. The values of $f(\mathrm{H}^+)$ at high energies agree with the results of Kreussler and Sizmann[10] who derived an empirical equation for the fraction of a proton beam exiting a carbon foil as neutrals, $f(\mathrm{H}^0)$, over an energy range 50 to 230 keV. At these energies, the fraction of $\mathrm{H}^-$ is negligible, so the exit fraction of $\mathrm{H}^+$ based on the results of Kreussler and Sizmann[10] is

$$f(\mathrm{H}^+) \approx 1 - f(\mathrm{H}^0) \qquad (5)$$

$$f(\mathrm{H}^+) \approx 1 - 0.01\ \exp(-1.07X^2 + 0.69X + 4.08)\ , \qquad (6)$$

where $X = \ln[1 + 0.7(v/v_0)^2]$, $v$ is the exit velocity of the transiting projectile, and $v_0$ is the Bohr velocity. Extrapolation of Eq. (6) to 30 keV yields $f(\mathrm{H}^+) = 0.395$, whereas Eq. (1) gives $f(\mathrm{H}^+) = 0.409$, showing good agreement between the extrapolated results of Kreussler and Sizmann and this study.

For helium, the fraction of the incident beam exiting as $\mathrm{He}^+$ increases approximately linearly with incident energy up to 30 keV. These results agree closely with the results of Burgi et al.[8] who examined the exit charge state distribution of He and O transiting nominal 1 to 6 $\mu$g cm$^{-2}$ carbon foils. No negative helium ions were observed in the exit charge state distributions.

For oxygen, both the exit charge state fractions of $\mathrm{O}^+$ and $\mathrm{O}^-$ increase with increasing energy, although the rate of increase for $f(\mathrm{O}^-)$ decreases with increasing energy [at energies higher than those used in this study, $f(\mathrm{O}^-)$ eventually decreases and equals 2.3% at 267 keV].[17] The fraction $f(\mathrm{O}^-)$ shows a large, random variation, which was similarly observed by Burgi et al.,[8] although their results for $f(\mathrm{O}^+)$ and $f(\mathrm{O}^-)$ were 40% higher than this study.

The effect on the exit charge state distributions of the exit surface chemistry, which could conceivably be manipulated to enhance the ionization yield, was investigated by depositing Al or Au layers on the exit surfaces of foils. Figure 3 shows a comparison of the exit charge state distribution $f(\mathrm{H}^+)$ from coated foils relative to uncoated foils as a function of incident proton energy. If we define $f_C$ as the fraction $f(\mathrm{H}^+)$ from uncoated (carbon) foils and $f_X$ as the charge state fraction $f(\mathrm{H}^+)$ from foils coated with either aluminum ($f_{\mathrm{Al}}$) or gold ($f_{\mathrm{Au}}$), then the variation between coated and uncoated foils is $(f_C - f_X)/f_C$. As shown in Fig. 3, the variation of the charge state distributions is less than approximately 2% and is not systematic. Therefore, the charge state distributions resulting from coated and uncoated foils are virtually identical, suggesting that the exit charge state distribution for incident hydrogen is independent of the exit surface chemistry modifications.

This effect is likely the result of foil history. The coated foils were exposed to air for approximately 1 d between their

Al or Au deposition and installation into the experimental chamber. This apparently results in surface contamination of the deposited layers (although physical and chemical growth kinetics of the deposition process, e.g., island growth, may also play an important role). Because all foils were "dirty," they exhibited identical charge state distributions. This has been previously observed in dirty foils,[10,12] and only "clean" foils with fresh surface layers deposited *in situ* showed slight variations in their charge state distributions.[10–12] Because foils used in a LENA sensor would be exposed to air at some point during the sensor assembly or calibration, we expect the foils to generate the exit charge state distributions equivalent to those depicted in Fig. 2 (although the cleaning effect of long-term, high vacuum exposure in space may slightly alter the distributions).

## 4 Angular Scatter Distributions

For a spinning spacecraft, the azimuthal angular resolution of a LENA imager would typically be fixed by a collimator, and the polar angular resolution would be a function of both the angular scattering of LENAs introduced by their interaction with the ultrathin foils (used for both charge stripping and pulse generation for coincidence or time-of-flight mass spectrometry) and the intrinsic position resolution of the detector.[7] Therefore, angular scattering can result in degradation of polar-angle information of a LENA's trajectory. In addition, large angular scattering in the charge-stripping foil can result in significantly reduced sensor throughput. Consequently, quantification of scattering of LENAs in a foil is vital in the design of a LENA imager.

A semiempirical theory describing multiple, low-angle scattering was derived by Meyer[18] and has in general been experimentally verified for foils thicker than used in this study.[19–21] We consider a projectile of atomic mass $m_1$ and atomic number $Z_1$ incident at an initial energy $E$ on a foil with atomic mass $m_2$, atomic number $Z_2$, thickness $t$, and atomic density $N$. The Meyer theory defines the reduced thickness ($\tau$), reduced scattering angle ($\Psi$), and reduced energy ($\varepsilon$) as

$$\tau=\pi a^2 N t \ , \tag{7}$$

$$\Psi=\psi\frac{\varepsilon}{2}\,\frac{m_1+m_2}{m_2} \ , \tag{8}$$

$$\varepsilon=\frac{aEm_2}{Z_1Z_2e^2(m_1+m_2)} \ , \tag{9}$$

where $e$ is the unit charge, $\psi$ is the scattering angle in the laboratory frame, and $a$, which is the Thomas-Fermi screening parameter, equals $0.885\ a_0(Z_1^{2/3}+Z_2^{2/3})^{-1/2}$, where $a_0$ is the Bohr radius (0.529 Å).

Based on the Meyer theory, a unique, energy-independent reduced halfwidth at half maximum $\Psi_{1/2}$ exists for each reduced foil thickness $\tau$, which is also energy independent. For example, for H transiting a 1.1 $\mu$g cm$^{-2}$ carbon foil, we obtain $\tau=0.93$ and $\Psi_{1/2}=0.28$ rad. Consequently, if we can determine $\Psi_{1/2}$ using the Meyer theory for a given projectile-foil combination, we can predict the scattering halfwidth $\psi_{1/2}$ in the laboratory frame from Eqs. (8) and (9):

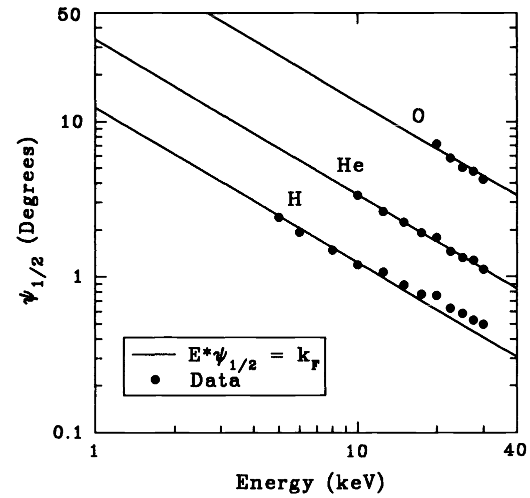



**Fig. 4** Angular scattering halfwidth at half maximum $\psi_{1/2}$ as a function of incident energy for H, He, and O transiting nominal 0.5 $\mu$g cm$^{-2}$ carbon foils. Solid line is fit to theory.

$$\psi_{1/2}E=k_F=\Psi_{1/2}\frac{2Z_1Z_2e^2}{a} \ . \tag{10}$$

The foil constant $k_F$ is unique for a particular projectile-foil combination and equals the product $\psi_{1/2}E$. Note that this equation illustrates an inverse proportionality between $\psi_{1/2}$ and the projectile energy for any given projectile-foil system.

Figure 4 shows the angular scattering halfwidth $\psi_{1/2}$ as a function of the incident energy for hydrogen, helium, and oxygen transiting nominal 0.5 $\mu$g cm$^{-2}$ carbon foils. The solid lines correspond to a least-squares fit of the data to the equation $k_F=\psi_{1/2}E$. The lines fit the data well, and the resulting values of the foil constant are $k_F=12.6$ keV deg for incident H, $k_F=33.6$ keV deg for incident He, and $k_F=133$ keV deg for incident O. Based on the Meyer theory described above, these values imply carbon foil thicknesses of 1.1, 1.5, and 1.9 $\mu$g cm$^{-2}$, respectively. The differences between the nominal foil thickness and the implied foil thicknesses can be attributed to greater foil thicknesses than cited by the manufacturer or thick layers of contamination. In fact, the thickness error cited by the manufacturer for nominal 0.5 $\mu$g cm$^{-2}$ foils is $\pm 0.5$ $\mu$g cm$^{-2}$, and foils may retain a significant amount of parting agent used in the manufacturing process. The systematic increase of implied foil thickness with increasing atomic mass of the projectile may be caused by actual thickness variations in the foils that were used or the inapplicability of the Meyer theory at these extreme foil thicknesses; this discrepancy is under current study.

Figure 5 depicts the angular scatter distributions $\psi$ for the $H^0$ and $H^+$ exit charge states from incident 20-keV $H^+$. The scatter distributions, which have been normalized to their charge states for comparison [e.g., the number of $H^+$ detected at a scattering angle $\psi$ divided by $f(H^+)$], are virtually indistinguishable. However, experiments of $H^+$ scattered from atomic targets (gases) show that the final charge state of incident $H^+$ undergoing an individual collision is a function of the scattering angle, so that charge exchange is dependent on the collision impact parameter.[22] Based on this, we might expect to observe a difference between the charge state distributions of $H^+$ and $H^0$ for small impact parameter colli-

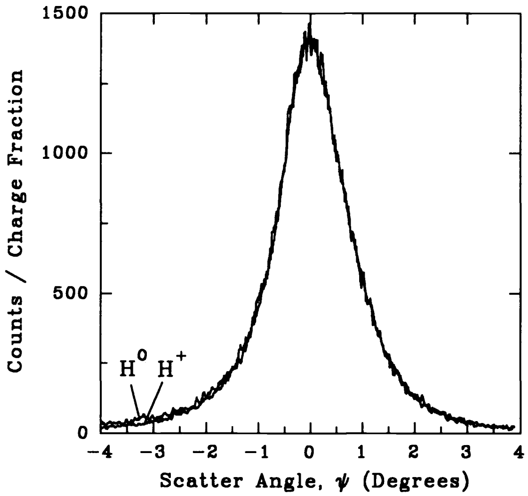



**Fig. 5** Angular scatter distributions of exiting $H^+$ and $H^0$ from 20-keV $H^+$ incident on a nominal 0.5 μg cm$^{-2}$ carbon foil. The distributions are normalized to the charge fraction for comparison.

sions, i.e., for $\psi \geq 1.5$ deg. Yet this feature is not observed in Fig. 5. Apparently, because a foil is composed of many atomic layers so that a projectile will experience multiple collisions before it exits the foil, enough collisions occur between a large-angle collision event and the exit surface of the foil so that an exiting projectile retains no information of the large-angle collision, i.e., charge state equilibration is established. Because no variation in the angular distribution for $H^+$ and $H^0$ is observed for large angle scatters (>1.5 deg) in Fig. 5, we therefore conclude that charge state equilibration occurs within a distance $\lambda$ that is much less than the foil thickness.

## 5 Applications to LENA Detection and Conclusions

Because (1) the dominant LENA species will typically be hydrogen, (2) the exit charge fraction of $H^+$ transiting carbon foils is large (5% at 1 keV to 41% at 30 keV), and (3) helium does not form negative ions, LENA instrumentation using a charge-stripping foil will typically detect a larger LENA flux, and therefore more statistically significant images, by detecting LENAs converted by the foil to positive ions rather than negative ions. Based on a minimum 2% ionization efficiency, LENA detection using conversion to positive ions is feasible for hydrogen at energies greater than ~1 keV, for helium greater than ~10 keV, and for oxygen greater than ~8 keV. However, because the exit charge fraction of $O^-$ is significantly greater than $O^+$, the detected flux of oxygen LENAs could be enhanced by configuring the imager to detect converted $O^-$, although this would result in detecting no He LENAs and detecting H LENAs at a low sensitivity.

Charge-stripping foils used in LENA sensors would be exposed to air during assembly and testing, and their surfaces will be contaminated. This study suggests that air contamination of Al and Au coated foils will result in identical charge state distributions. Consequently, coating the foil will not enhance its ionization efficiency.

Significant angular scattering of LENAs in the carbon foil can result in loss of information of their incident trajectories (reduced polar resolution of a LENA imager) or scattering into instrument walls, which prevents their detection (degraded sensitivity). Consequently, knowledge of the scatter distribution is required to define imager specifications. Scatter distributions were quantified by deriving a foil constant that equals the product of the angular scatter halfwidth $\psi_{1/2}$ and the incident projectile energy. Significant differences in the scattering halfwidth between the empirical results of this study and the theory of Meyer[18] are observed and are likely caused by foil thicknesses larger than those cited by the manufacturer. Based on the results presented here, the scattering halfwidth $\psi_{1/2}$ ranges from 2 deg at 5 keV to 0.5 deg at 30 keV for H, from 6.6 deg at 5 keV to 1.1 deg at 30 keV for He, and from 27 deg at 5 keV to 4.4 deg at 30 keV for O. Because of the large scattering halfwidth for oxygen, high angular resolution of incident neutral oxygen may require use of even thinner (0.2 to 0.3 μg cm$^{-2}$) foils, which are currently under study.


### Acknowledgments

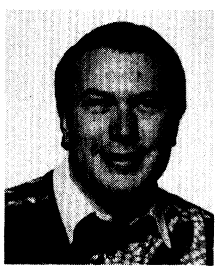
We gratefully acknowledge J. Baldonado and D. Everett for their laboratory support and D. Deck for Al/Au deposition. This work was performed under the auspices of the United States Department of Energy.


### References


1. E. C. Roelof, "Energetic neutral atom image from a storm-time ring current," *Geophys. Res. Lett.* **14,** 652–655 (1987).
2. D. J. McComas, B. L. Barraclough, R. C. Elphic, H. O. Funsten, and M. F. Thomsen, "Magnetospheric imaging with low-energy neutral atoms," *Proc. Natl. Acad. Sci. U.S.A.* **88,** 9598–9602 (1991).
3. K. R. Moore, D. J. McComas, H. O. Funsten, and M. F. Thomsen, "Low energy neutral atoms in the earth's magnetosphere: modeling," *Proc. SPIE* **1744,** 51–61 (1992).
4. R. W. McEntire and D. G. Mitchell, "Instrumentation for global magnetospheric imaging via energetic neutral atoms," *Geophys. Monogr. Ser.* **54,** 69–80 (1989).
5. K. C. Hsieh and C. C. Curtis, "Remote sensing of planetary magnetospheres: Mass and energy analysis of energetic neutral atoms," *Geophys. Monogr. Ser.* **54,** 159–164 (1989).
6. K. C. Hsieh, E. Keppler, and G. Schmidtke, "Extreme ultraviolet induced forward photoemission from thin carbon foils," *J. Appl. Phys.* **51,** 2242–2246 (1980).
7. D. J. McComas, H. O. Funsten, J. T. Gosling, K. R. Moore, and M. F. Thomsen, "Low energy neutral imaging," *Proc. SPIE* **1744,** 40–50 (1992).
8. A. Burgi, M. Oetliker, P. Bochsler, and J. Geiss, "Charge exchange of low energy ions in thin carbon foils," *J. Appl. Phys.* **68,** 2547–2554 (1990).
9. H. O. Funsten, D. J. McComas, and B. L. Barraclough, "Thickness uniformity and pinhole density analysis of thin carbon foils using incident keV ions," *Nucl. Instrum. Methods* **B66,** 470–478 (1992).
10. S. Kreussler and R. Sizmann, "Neutralization of 50–230 keV hydrogen ions which have penetrated Al, Au, C, and Cs films," *Phys. Rev. B* **26,** 520–529 (1982).
11. R. Behrisch, W. Eckstein, P. Meischner, B. M. U. Scherzer, and H. Verbeek, "Charged fraction of 5 keV to 150 keV hydrogen atoms after emergence from different metal surfaces," in *Atomic Collisions in Solids,* S. Datz, B. R. Appleton, and C. D. Moak, Eds., Vol. 1, pp. 315–329, Plenum Press, New York (1975).
12. K. H. Berkner, I. Bornstein, R. V. Pyle, and J. W. Stearns, "Charge fractions and excited-atom populations of 8–100 keV deuterium beams emerging from solid films of C, Mg, Nb, and Au," *Phys. Rev. A* **6,** 278–288 (1972).
13. H. D. Betz, "Heavy ion charge states," in *Applied Atomic Collision Physics,* S. Datz, Ed., pp. 1–42, Academic Press, New York (1983).
14. M. Oetliker, "Charge state distribution, scattering and residual energy of ions after passing through thin carbon foils," Thesis, University of Bern, Switzerland (1989).
15. H. O. Funsten, B. L. Barraclough, and D. J. McComas, "Shell effects observed in exit charge state distributions of 1–30 keV atomic projectiles transiting ultra-thin carbon foils," *Nucl. Instrum. Methods B* **80/81,** 49–52(1993).

16. J. F. Ziegler, "TRIM computer code," IBM-Research, Yorktown, N.Y. (1992).
17. W. N. Lennard, D. Phillips, and D. A. S. Walker, "Equilibrium charge state distributions of ion beams exiting carbon foils," *Nucl. Instrum. Methods* **179,** 413–419 (1981).
18. L. Meyer, "Plural and multiple scattering of low-energy heavy particles in solids," *Phys. Status Solidi B* **44,** 253–268 (1971).
19. F. Bernhard, P. Krygel, R. Manns, and S. Schwabe, "Multiple scattering of various ions in carbon and germanium," *Radiat. Eff.* **13,** 249–255 (1972).
20. H. H. Andersen and J. Bøttiger, "Multiple scattering of heavy ions of keV energies transmitted through thin carbon foils," *Phys. Rev. B* **4,** 2105–2111 (1971).
21. G. Högberg, H. Nordén, and H. G. Berry, "Angular distributions of ions scattered in thin carbon foils," *Nucl. Instrum. Methods* **90,** 283–288 (1970).
22. D. Eichenauer, N. Grün, and W. Scheid, "Classical trajectory calculations for $H^+ - H$ collisions with the Wigner function as initial phase space distribution," *J. Phys.* **14,** 3929–3941 (1981).

**Herbert O. Funsten** received a BS from Washington and Lee University and an MEP and a PhD in engineering physics from the University of Virginia. He received a postdoctoral fellowship at Los Alamos National Laboratory (LANL) in 1990, and in 1993 became a technical staff member at LANL in the Space and Atmospheric Sciences Group. His research interests include ion and electron beam interactions with solids, ion- and electron-beam-induced film deposition, space-plasma interactions with the Moon, and space-plasma and neutral atom spectrometer design and development. He is the author or coauthor of over 25 technical papers.



**David J. McComas** is the group leader for Space and Atmospheric Sciences at Los Alamos National Laboratory. Dr. McComas is a fellow of the American Geophysical Union and a recipient of its James B. Macelwane Award. Dr. McComas is the principle investigator for the Department of Energy's (DOE's) Magnetospheric Plasma Analyzer instruments and is coinvestigator on several National Aeronautics and Space Administration (NASA) programs including the plasma instrument for Cassini, the Thermal Ion Dynamics Experiment on the International Solar Terrestrial Physics (ISTP) Polar spacecraft, the Advanced Composition Explorer solar wind spectrometers, and the Ulysses solar wind spectrometers. Dr. McComas presently serves on the National Research Council's Committee on Solar Terrestrial Research and NASA's Inner Magnetosphere Imager study team. Dr. McComas is an associate editor for the *Journal of Geophysical Research (Space Physics)* and is an author of over 100 scientific papers on topics ranging from space instrument design to solar wind, magnetospheric, cometary, and planetary physics.

**Bruce L. Barraclough** has been at Los Alamos National Laboratory since 1978 and is presently a technical staff member with the Space and Atmospheric Sciences Group. He received his undergraduate degree in chemistry and geology from the University of New Mexico and his graduate degree in geochemistry from the University of California at Los Angeles. He is currently involved in all aspects of designing, fabricating, and flying plasma and neutron sensors for spaceflight and in analyzing the returned scientific data. In addition to being the principal investigator for a series of neutron detectors currently aboard a constellation of geosynchronous satellites and for an additional follow-on series, he is coinvestigator or otherwise involved with plasma and neutron instruments aboard Cassini, Cluster, Ulysses, Combined Release and Radiation Effects Satellite (CRRES), and several other National Aeronautics and Space Administration and Department of Defense/Department of Energy missions. His current interests range form instrument design and secondary electron/particle physics to solar neutron emission.